FoodWise AI — Deep Learning Training & Evaluation Report (V2)

Faculty Evaluation Report | Amrita Vishwa Vidyapeetham | Generated: 2026-10-12 00:01:24

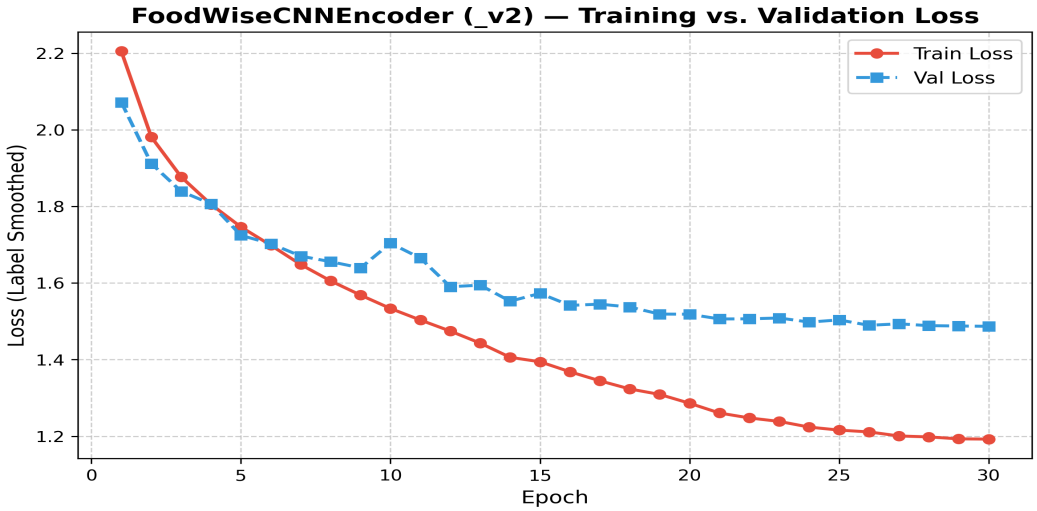
1. Executive Summary & Overview

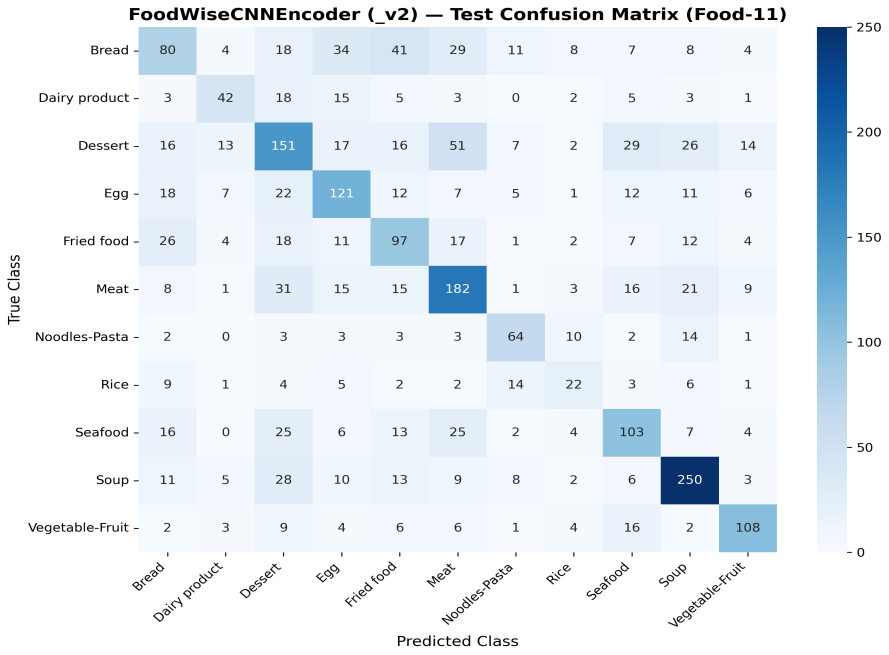
FoodWise AI evaluates three distinct PyTorch neural network models for visual appetite recommendations. This extended series (V2) evaluates the effect of 30 training epochs with cosine annealing schedule and label smoothing on Food-11, paired with 25-epoch evaluations of the Metabolic Autoencoder and NCF Matcher. All model checkpoints and outputs are uniquely saved under version suffix '_v2'.

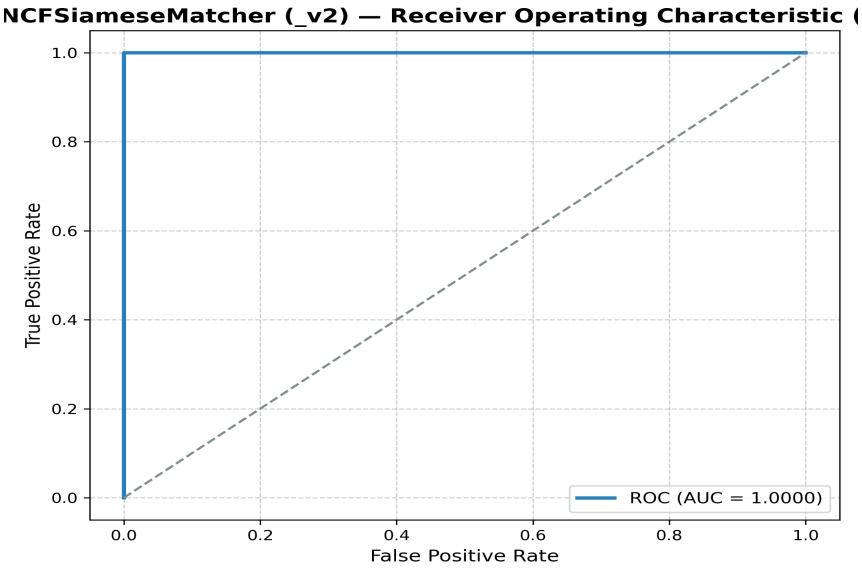
2. Model Performance Summary

Model	Dataset / Source	Best Epoch	Primary Metric	Secondary Metric	Training Time
FoodWiseCNNEncoder_v2	Food-11 (7,862 train / 2,291 test)	Epoch 30	Acc: 53.25%	Macro F1: 0.5153	1995.5s
Profile Autoencoder_v2	Synthetic Profiles (5,000 vectors)	Epoch 25	MSE: 0.000826	MAE: 0.0205	3.9s
NCF Matcher_v2	Synthetic Pairs (4,000 pairs)	Epoch 1	Acc: 100.00%	ROC-AUC: 1.0000	4.8s

3. Selected Evaluation Figures




FoodWiseCNNEncoder (_v2) — Test Confusion Matrix (Food-11)
True Class
Predicted Class
Bread
Dairy product
Dessert
Egg
Fried food
Meat
Noodles-Pasta
Rice
Seafood
Soup
Vegetable-Fruit
250
200
150
100
50
0


NCFSiameseMatcher (_v2) — Receiver Operating Characteristic (
True Positive Rate
False Positive Rate
ROC (AUC = 1.0000)
0.0
0.2
0.4
0.6
0.8
1.0